



YISHUN INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
Higher 1

NAME

ANSWERS

INDEX NO

CG

BIOLOGY

8876/02

Paper 2 Structured Questions and Free Response Questions

1 Sep 2021

2 hours

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name, index no. and CG on this cover page.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question paper.

Section B

Answer any **one** question in the spaces provided on the Question Paper. Indicate the question you have attempted at the top of page **18**.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

Overall Results for JC2 H1 Biology Preliminary Examination				
Paper	Raw Marks	Weighting	%	Grade
1	30			
2	60			

For Examiner's Use	
Section A	
1	7
2	15
3	13
4	10
Section B	
	15
Total	60

This document consists of **19** printed pages and **1** blank page.

Section A

Answer **all** questions in this section.

- 1 To investigate the amount of time cells spent in each phase of the cell cycle, a population of dividing diploid cells is stained with a dye that becomes fluorescent when it binds to DNA. The amount of fluorescence is directly proportional to the amount of DNA in each diploid cell.

To measure the amount of DNA in each cell, the population of dividing diploid cells are passed through a fluorescence-activated cell sorter (FACS), an instrument that measures the amount of fluorescence in individual cells. The number of cells with a given DNA content, in each phase of the cell cycle, is plotted on a graph shown in Fig. 1.1.

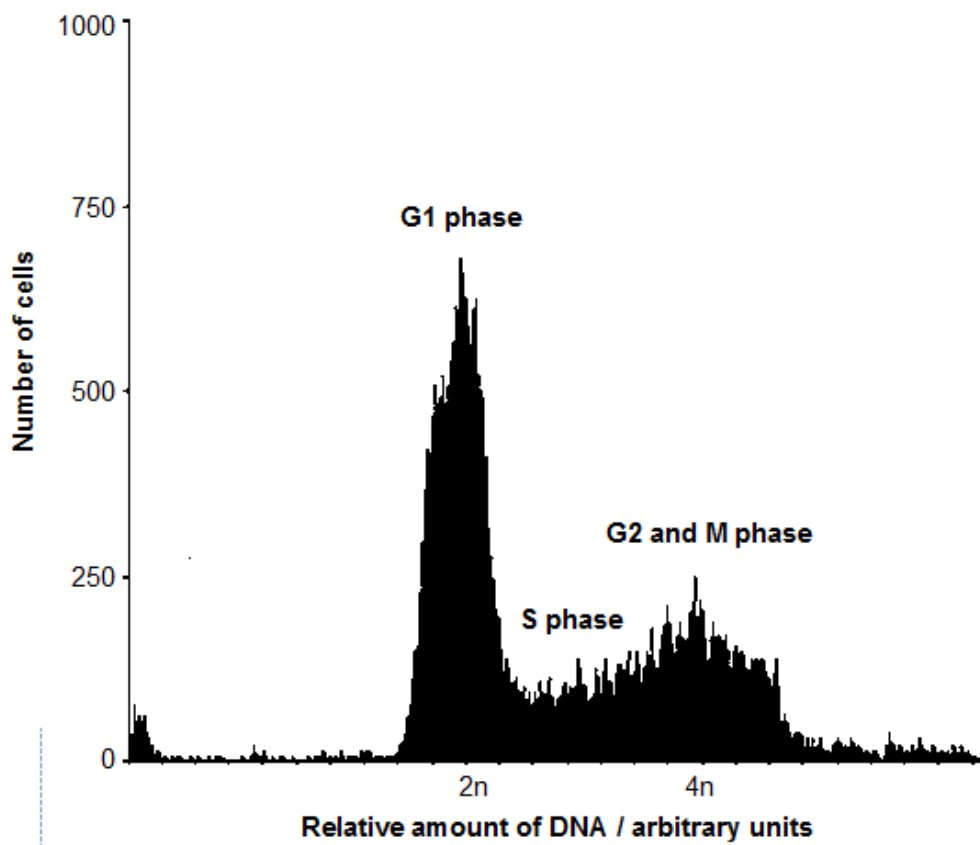


Fig. 1.1

(a) With reference to Fig. 1.1,

- (i) explain how the relative amount of DNA changes as the cells progress through the different phases of the cell cycle.

1. In G1 phase, the diploid cells contain $2n$ arbitrary units of DNA,

as the cells are preparing for nuclear division;

2. As the cells progressed from G1 to S phase, DNA replication occurs in S phase,

thus increasing the amount of DNA in the cells to more than $2n$ arbitrary units;

3. DNA replication is completed at the end of S phase,

the cells contain twice the amount of DNA, $4n$ arbitrary units, in G2 and M phase;

[3]

- (ii) suggest why the G1 phase is the longest phase of the cell cycle in this population of cells.

G1 phase has the highest number of cells;

[1]

(b) Cell division is *almost* identical in animal and plant cells.

Describe three differences in cell division between an animal cell and a plant cell.

1. Animal cell contains a pair of centrioles (which replicates and migrates to opposite poles of the dividing cell,

but centrioles are absent in plant cell;

2. Animal cell's centrioles develops into microtubules organising centre / develop into asters,

3. but asters are absent in plant cell;

Animal cell forms cleavage furrow to divide into two genetically identical daughter cells,

4. but plant cell forms a new cell wall / cell plate to divide into two genetically identical daughter cell;

Animal cell are completely separated into two daughter cells;

in plants cells, the two daughter cells are joined by middle lamella;

[3]

Any 3 points above

[Total: 7]

- 2 A key reaction in photosynthesis occurs when ribulose biphosphate carboxylase (Rubisco) catalyses the fixation of carbon dioxide to ribulose biphosphate (RuBP).

Fig. 2.1 shows the summary of events occurring during the translation of Rubisco mature mRNA.

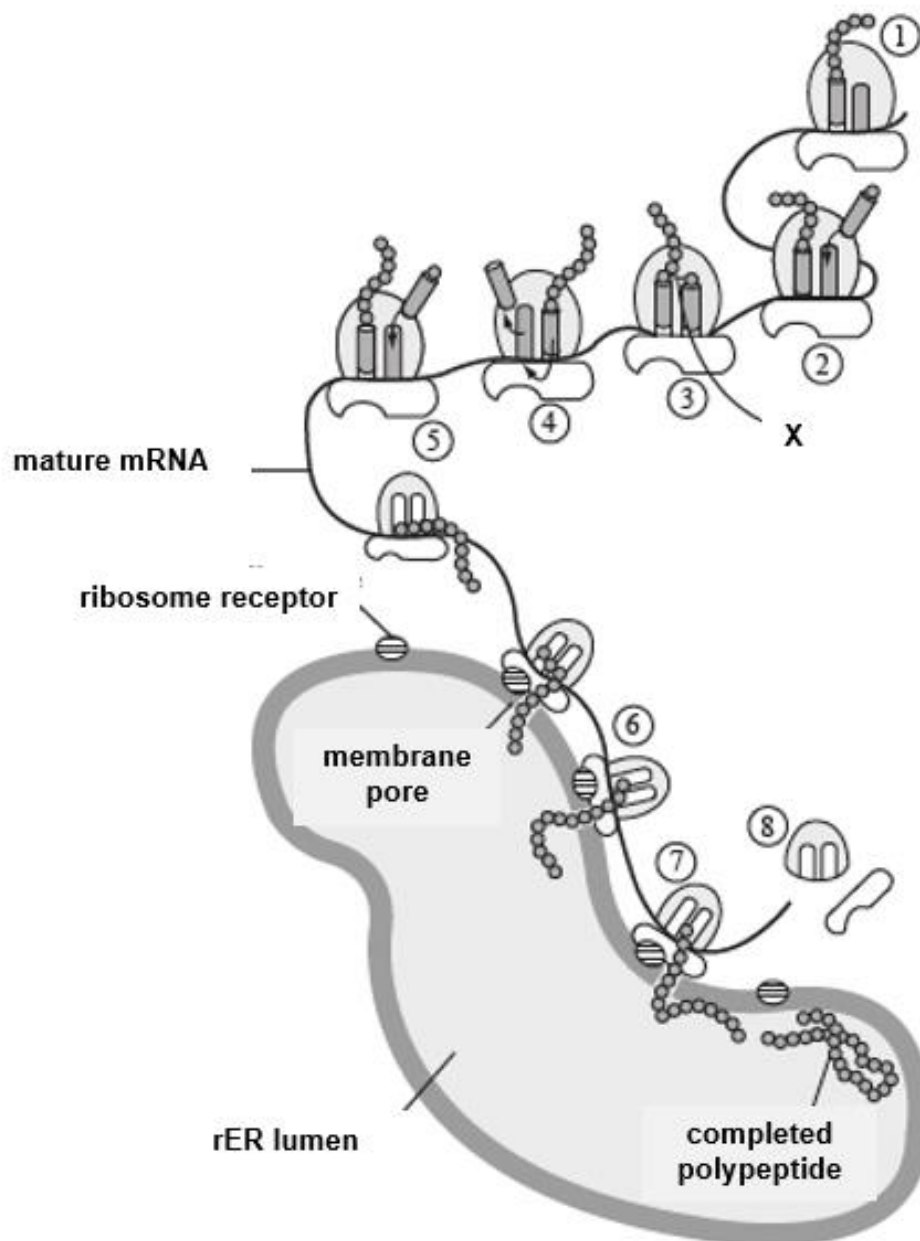


Fig. 2.1

- (a) With reference to Fig. 2.1,
- (i) name the bond labelled X.
- Peptide bond;**

[1]

(ii) outline the processes occurring in stages 4 and 5.

1. [Stage 4] tRNA in E site detaches / leaves the ribosome,

Ribosome moves by one codon along mature mRNA, translocating aminoacyl tRNA complex in A site to E site;

2. [Stage 5] aminoacyl tRNA complex in P site carries the growing polypeptide chain;

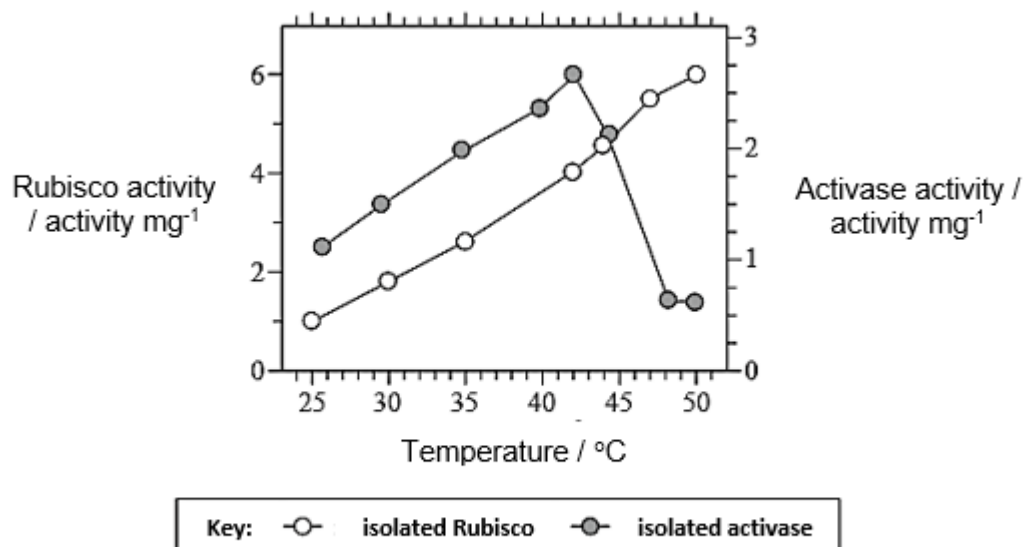
another aminoacyl tRNA / tRNA carrying an amino acid enters the A site of the large ribosomal subunit of ribosome;

Reject: tRNA moves to P site

[2]

To be effective, Rubisco must be activated by another enzyme called activase. The activities of Rubisco and activase, each isolated from tobacco leaves, were independently investigated in a laboratory, under conditions of increasing temperature.

Fig. 2.2 shows the effect of temperature on the activities of the two enzymes.



[Source: adapted from S. Crafts-Brandner and M. Salvucci (2000) 'Rubisco activase constrains the photosynthetic potential of leaves at high temperature and CO₂.' *PNAS*, 97, pp. 13 430–13 435. Figure 2.]

Fig. 2.2.

(b) With reference to Fig. 2.2,

(i) describe the relationship between Rubisco activity and temperature.

As temperature increases from 25°C to 50°C,

Rubisco activity increases from 1 activity mg⁻¹ to 6 activity mg⁻¹;

[1]

(ii) explain the change in activase activity at temperatures higher than 42 °C.

1. [Reference to Fig. 2.1] As temperature rises beyond the optimal temperature 42°C where activase activity is optimum at 2.75 activity mg^{-1} ;

activase activity drops from 2.75 activity mg^{-1} to ~0.6 activity mg^{-1} at 48 °C,

2. [Explain] Increase in kinetic energy leads to thermal agitation of activase, resulting in breakage of bonds such as hydrophobic interactions and H bonds,

leading to unfolding of secondary and tertiary structure of activase;

3. loss of specific 3D conformation of active site, $\therefore$ no longer complementary in shape, size, charge and orientation to substrate,

$\therefore$ unable to form enzyme-substrate complexes, activase is denatured;

[3]

(c) In a leaf, both enzymes are present together. Predict, with a reason, how the rate of photosynthesis would change from 35°C to 50°C.

1. As temperature increases from 35°C to 42°C, there is an initial increase in photosynthesis rate up to 42°C,

followed by a decrease in photosynthesis rate after 42°C;

2. Beyond 42°C, decreased activity of activase lowers the activation of Rubisco, leading to less carbon fixation (to RuBP) in the light-independent reaction / Calvin cycle $\rightarrow \therefore$ decreasing rate of photosynthesis;

[2]

(d) Light is not a factor that directly limits the rate of the Calvin cycle but it is needed for the Calvin cycle to continue.

Outline why light is needed for the Calvin cycle to continue.

1. Light is needed (in the light-dependent reactions) for the synthesis of ATP

and reduced NADP;

2. ATP serves as the energy source for the Calvin cycle and is needed to

phosphorylate glycerate 3-phosphate (GP) to glycerate biphosphate / to regenerate RuBP;

OR

3. reduced NADP provides the reducing power for the Calvin cycle and

is needed to reduce glycerate biphosphate to triose phosphate (TP) (which is required for the synthesis of sugars and regeneration of RuBP);

[2]

In plants, the carbohydrates manufactured in photosynthesis are used to form polysaccharides such as starch.

(e) Explain how the structure of starch is adapted for its function in plants.

Any 2 structural features and corresponding functions:

1. Consists of thousands of glucose monomers,
2. Hydrolysis of starch by enzymes e.g. amylase releases many glucose monomers, which can be used as a respiratory substrate for ATP synthesis in cellular respiration;
3. Helical structure of amylose and branched structure of amylopectin makes starch compact,
4. allows efficient storage of glucose molecules i.e. more molecules of starch can be stored within less space in plant cells;
5. α -glycosidic bonds between glucose monomers,
6. can be broken down by amylase to release glucose molecules for respiration;
7. Folding of the starch molecule such that the hydroxyl groups (-OH) are in the interior of the molecule, i.e. prevented from interacting with water,

OR

Large molecular weight of starch,

8. Makes starch insoluble (inability to form hydrogen bonds with water molecules), allowing it to be stored without changing the osmotic potential within the plant cell;
9. Branched structure of amylopectin,
10. allows multiple action sites for enzymatic action by amylase, increasing the rate of hydrolysis to release glucose monomers for respiration;

[4]

[Total: 15]

3 There have been many breakthroughs in stem cell research in recent years. It has been discovered that stem cells are involved in the replacement of worn-out cells and repair of damaged tissues. Further research is being conducted to better understand the mechanism involved in controlling the behaviour of stem cells in order to better manipulate them to treat various diseases and disorders.

(a) (i) State the type of stem cells involved in the replacement of worn-out cells and repair of damaged tissues

Adult stem cells;

[1]

(ii) Describe **two** unique properties of this type of stem cells.

Any 2 below:

1. Undifferentiated cells (found in differentiated tissues)

do not possess tissue specific cell structures to perform specialised function;

2. Multipotent,

able to differentiate into a limited range of cell types;

3. Able to undergo mitotic cell division for self-renewal; OR

Able to undergo asymmetric division to give rise to a stem cell & a progenitor cell,

which further divides & differentiates into specialised cell types to replace worn out / damaged cells;

[2]

(b) Explain the importance of mitosis in the repair of damaged tissue.

produces genetically identical daughter cells ; *Reject: genetically similar cells*

to replace to play the same function; (*reject: repair damaged cells*)

[1]

(c) Induced pluripotent stem cells (iPSCs) are promising tools for cell replacement therapy. This is a medical procedure in which living iPSCs are injected into a patient to treat certain genetic diseases.

State **one** advantage of using iPSCs for cell replacement therapy.

Any 1 below:

1. iPS cells are derived from adult cells of the patient, hence no immune rejection; (*Reject: ref. to tissue rejection*)

2. Using iPS cells can overcome ethical issues of destroying human embryos to obtain pluripotent stem cells, since iPS cells are derived from adult cells;

[1]

- (d) Scientists have detected telomerase in more than 90% of human tumour samples. Stem cells also express high levels of telomerase and can divide indefinitely.

Explain why normal adult stem cells do not give rise to tumour, despite having a high level of telomerase activity.

1. Cell division in normal adult stem cells can be regulated;

OR

Stem cells will only divide when necessary e.g. in the presence of growth factors;

2. As they obey cell cycle control/ stop appropriately at cell cycle checkpoints when conditions at previous stage are not met;

OR

arrest of cell cycle to repair DNA damage / cells send to apoptosis when DNA damage is irreparable;

[2]

The appearance of cancer cells has been known to be different from normal cells. These differences have been used as a method of diagnosis by doctors. Prostate cells taken from wild-type mice and mice with prostate cancer were analysed.

Fig. 3.1 compares the same organelle found in these cells viewed under the electron microscope, while Fig. 3.2 shows the levels of ribosomal RNA (rRNA) measured in these two types of cells.

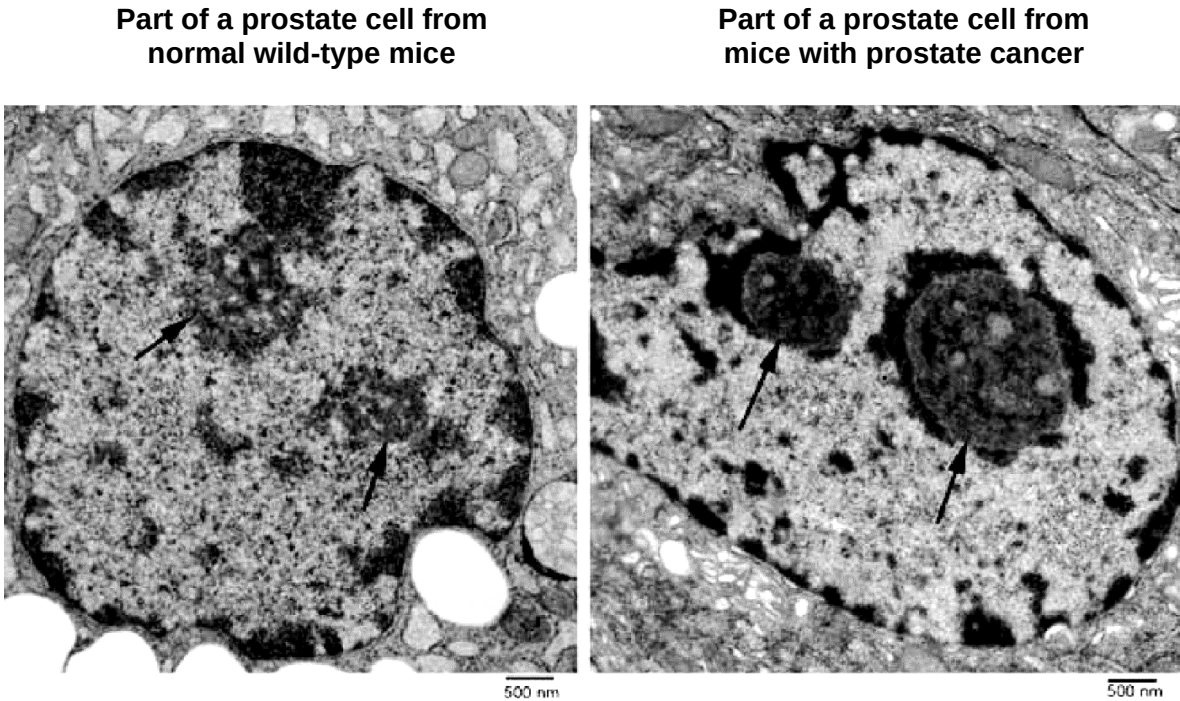


Fig. 3.1

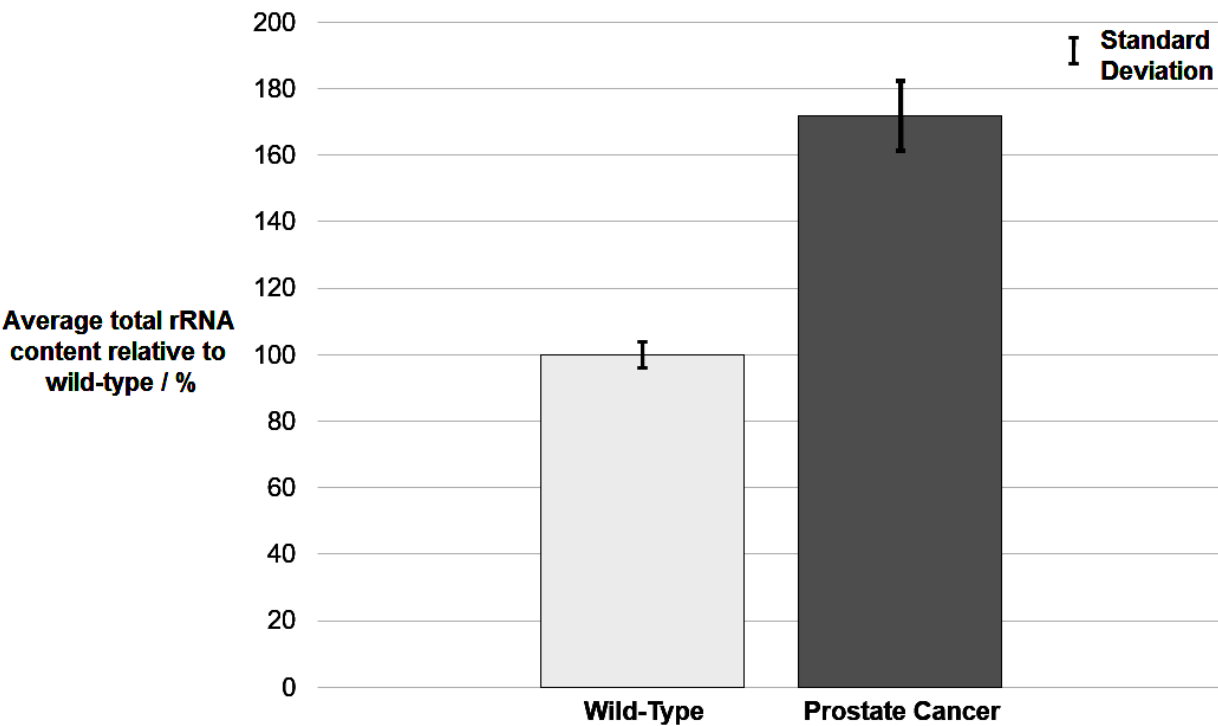


Fig. 3.2

- (e) (i) Identify the regions indicated by the arrows in Fig. 3.1.

Nucleoli; (Reject: nucleolus, nucleus)

[1]

- (i) With reference to the regions identified in Fig. 3.1 and information given in Fig. 3.2, describe the differences between normal prostate cells and prostate cancer cells.

Figure 3.1:

1. Prostate cancer cells have denser / larger nucleoli;

Figure 3.2:

2. Prostate cancer cells contain more rRNA,

with 72% higher in average total rRNA content / 170% vs. 100 % in normal prostate cells;

OR

3. Prostate cancer cells show greater variation in rRNA content compared to normal prostate cells,

higher standard deviation of 20% vs ~ 8% in normal prostate cells;

[2]

- (iii) Explain the differences described in (e)(ii).

Any 3 points below:

1. Nucleoli are the sites of rRNA synthesis / sites of transcription of rRNA genes/

are the sites of assembly of ribosomal subunits, hence appear denser in cancer cells;

2. Cancer cells undergo higher rates of protein synthesis in preparation for faster rate of cell division;

3. Higher rates of translation is facilitated by a greater number of ribosomes, hence the higher rRNA content in cancer cells;

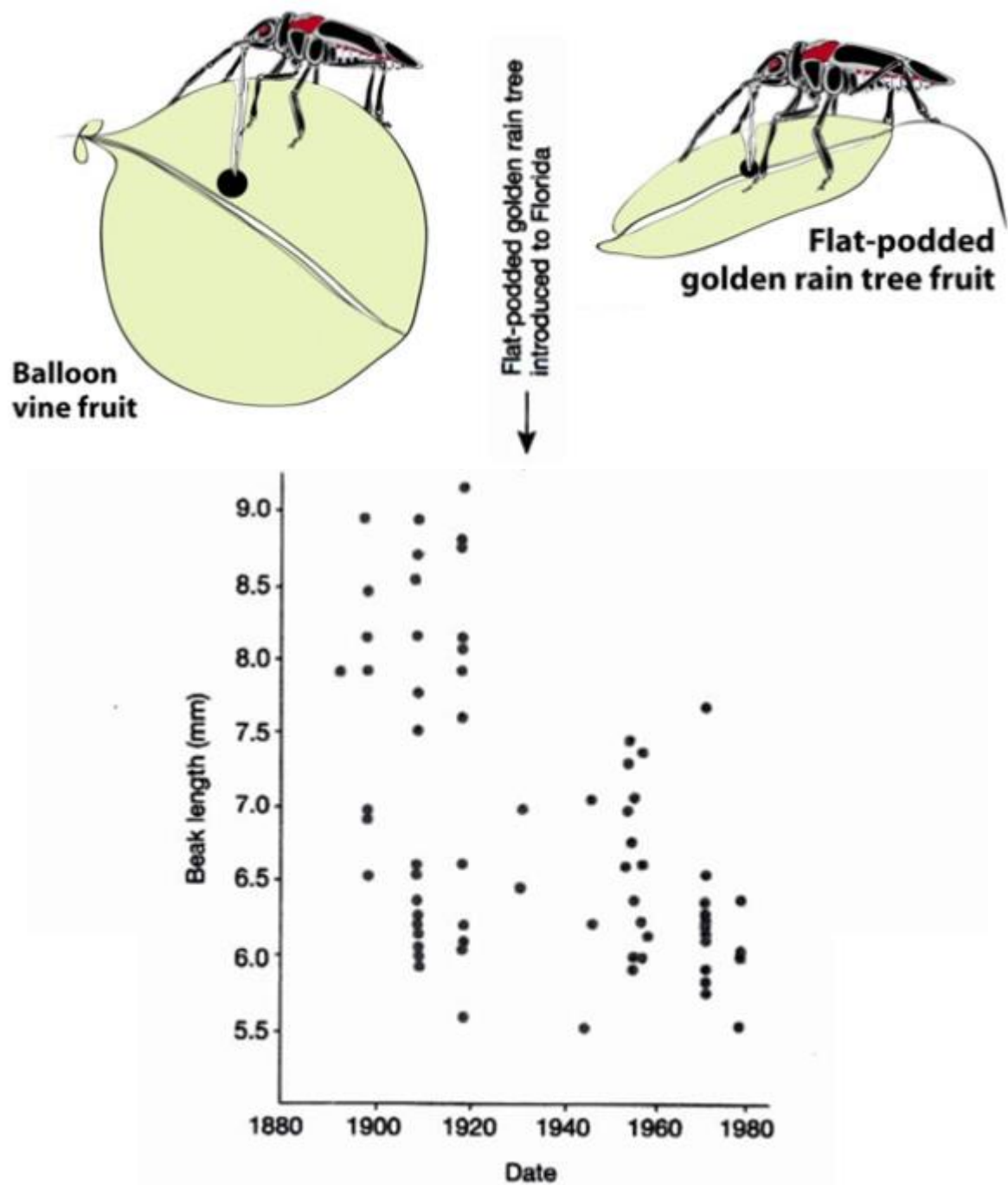
4. (Different cancer cells have different mutations resulting in) dysregulation of gene expression in cancer cells lead to different cells having different rates of protein synthesis and greater variation in rRNA content;

[3]

Accept reverse argument

[Total: 13]

- 4 Soapberry bugs in Florida originally fed on the native balloon vine, using their sharp beaks to penetrate the fruits to reach the seeds. After the introduction of the flat-podded golden rain tree from Asia to mainland central Florida, the phenotype of the soapberry bug has changed, as shown in Fig. 4.1.



Legend: Each dot on the graph shows the average beak length of 1000 bugs of a population.

Fig. 4.1

- (a) With reference to Fig. 4.1, describe how the phenotype of the soapberry bugs has changed after the introduction of the flat-podded golden rain tree fruit in 1930.

1. Soapberry bug's beak length has reduced from a range of 5.6 - 9.1mm between 1885 to 1930,

to a range of 5.5 - 7.6mm after the introduction of the fat-podded golden rain tree fruit in 1930;

[1]

- (b) The flat-podded golden rain tree fruit has thinner-skinned fruit compared to that of balloon vine fruit.

Explain how this has resulted in the change in soapberry bug's phenotype after 1930.

1. [Idea of pre-requisite for natural selection] There is genotypic variation that leads to

phenotypic variation in beak length which exist in the soapberry bug population;

2. [Identify selection pressure] The introduction of the flat-podded golden rain tree fruit as an alternative food source acts as a new selection pressure,

selecting for individuals with favourable phenotypes of shorter beak lengths in soapberry bug population;

3. [Natural selection as agent that drives evolution] soapberry bugs with shorter beak lengths giving them better access to seeds found nearer to surface of thinner-skinned fruit are at a selective advantage

and have higher survival rates and greater reproductive success,

passing on their advantageous alleles coding for shorter beak length to their offspring;

4. [Idea of evolution of population] resulting in a change in allele frequency in the gene pool of the population / Higher proportional of alleles coding for shorter beaks in the gene pool over time,

→ overall decrease in beak length after 1930;

[4]

- (c) Describe **one** source of genetic variation accounting for the differences in beak lengths in the soapberry bugs.

Any 1 below:

1. crossing over between non-sister chromatids of homologous chromosomes and independent assortment in meiosis in F_1 ,

give rise to different combinations of alleles in gametes;

2. independent arrangement / assortment of non-identical sister chromatids at the metaphase plate during metaphase II and subsequent segregation / separation of non-identical sister chromatids during anaphase II,

results in gametes with numerous combinations of maternal & paternal chromosomes;

3. random fusion of gametes;

4. mutation in gene coding for beak length;

[1]

- (d) In soapberry bugs, a recessive mutant allele on chromosome 2 leads to vestigial wings as opposed to the normal wing size.

A second eyeless gene on the X chromosome encodes for a transcription factor required for normal eye development. A recessive mutant allele at this gene locus leads to absence of eye structures as opposed to normal eyes.

A female fly heterozygous at both gene loci was crossed with a normal eyed male fly heterozygous for wing size.

Using the following symbols,

X^E : dominant allele on X chromosome for normal eyes

X^e : Recessive allele on X chromosome for eyeless

A: dominant allele for normal wing size

a: recessive allele for vestigial wings

draw a genetic diagram to show the possible offspring of the cross.

Parental phenotype:

Normal wing size,
normal-eyed male

Normal wing size,
normal-eyed female

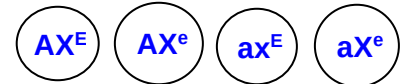
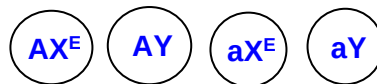
Parental genotype
[1]:

AaX^EY

×

AaX^EX^e

Gametes [1]:



Fertilisation [1]:

Gametes	AX^E	AX^e	aX^E	aX^e
AX^E	$AA X^E X^E$	$AA X^E X^e$	$Aa X^E X^E$	$Aa X^E X^e$
AY	$AA X^E Y$	$AA X^e Y$	$Aa X^E Y$	$Aa X^e Y$
aX^E	$Aa X^E X^E$	$Aa X^E X^e$	$aa X^E X^E$	$aa X^E X^e$
aY	$Aa X^E Y$	$Aa X^e Y$	$aa X^E Y$	$aa X^e Y$

F₁ genotypic ratio:

$3 A_X^E X^E :$
 $3 A_X^E X^e :$

$1 aa X^E X^E :$
 $1 aa X^E X^e$

$3 A_X^E Y :$

$3 A_X^e Y :$

$1 aa X^E Y :$

$1 aa X^e Y$

F₁ phenotypic ratio:

6 :
Normal wing size,
normal-eyed female

2 :
vestigial wings,
normal-eyed female

3 :
Normal wing size,
normal-eyed male

3 :
Normal wing size,
eyeless male

1 :
Vestigial wings,
normal-eyed male

1
Vestigial wings,
eyeless male

Correct F₁ genotypic and corresponding phenotypic ratio [1]

[4]

[Total: 10]

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

5 (a) Describe how a gene mutation can lead to a different protein product. [6]

1. Definition of gene mutation: Change in base sequence that leads to a change in amino acid sequence and eventually change in phenotype.
2. Usually involve a point mutation such as insertion, deletion or substitution of a nucleotide.
3. Insertion/deletion of a nucleotide leads to frameshift mutation / results in a change in reading frame.
4. Results in a change in codons and hence amino acid sequence from point of insertion / deletion onwards
5. Substitution on one nucleotide leads to missense mutation that involves a change in one codon in the mRNA
6. that codes for a different amino acid which has a different R group.
7. Both frameshift and missense mutation will change the primary structure due to the presence of different R groups.
8. Folding of the polypeptide chain to tertiary structure will change/be different due to different bonds between R groups.
9. Eventually leading the gene product having a different 3D conformation/tertiary structure that causes a loss of its function.
10. Nonsense mutation leading to a premature stop codon leads to a truncated protein / incomplete polypeptide chain which also fold into a non-functional protein.
11. Illustrate with a valid example [max 3 if student only write about example with no reference to other form of mutations]

Example: sickle cell anaemia

1. Single base substitution: GTG for GAG on the β -globin gene
2. Leads to translation of glutamine (which possesses polar R group) to valine (with hydrophobic R groups) → change in primary structure of β -globin
3. Results in a hydrophobic R group which projects out of the globular protein
4. At low oxygen pressure → binds to other haemoglobin chains via hydrophobic interactions
5. Haemoglobin aggregates to form long fibres which leads to sickle shaped red blood cells

QWC [1]:

At least 2 different types of gene mutation leading to change in protein structure

- (b) The expression of genes gives rise to proteins that affect the biochemical reactions and other processes in organisms.

Discuss, with examples, the importance of specific shapes of proteins in organisms.

[9]

1. Active site of enzyme has specific shape that substrate can fit into
2. Via lock and key mechanism
3. **[Importance]** To form enzyme-substrate complex/ products important for metabolic pathways / increase the rate of reaction
4. **[Example]** any enzyme and substrate

5. DNA to fit into binding site of proteins (point 6-9, 16-19)
6. **[Importance]** Ref. to DNA replication
7. **[Example]** DNA polymerase binds to 3' OH group of existing nucleotide/ helicase binds to origin of replication/ topoisomerase binds to DNA to unzip the double-stranded
8. **[Importance]** Ref. to transcription
9. **[Example]** transcription factor binding to DNA / RNA polymerase binds to promoter

10. Binding of substances to transport proteins
11. **[Importance]** Allows for movement of substances across cell membrane
12. **[Example]** transmembrane protein e.g. Na⁺ channel, Na⁺-K⁺ pump, glucose transporter etc.

13. **[Example]** Haemoglobin is made up of 4 polypeptides and their haem groups to form a specific conformation
14. **[Importance]** allows it to bind to oxygen molecules to form oxyhaemoglobin/ transport oxygen to all parts of the body
15. Reference to cooperative binding

16. **[Importance]** Ref. to separation of sister chromatids during anaphase
17. E.g. kinetochore to bind to centromere via complementary shape
18. **[Importance]** Ref. to amino acid activation
19. E.g. tRNA anticodon bind to the anticodon attachment site of amino-acyl tRNA synthetase

QWC [1]

Importance of specific shapes communicated accurately and to include at least two examples

- 6 (a) Describe the role of membranes in photosynthesis and cellular respiration.

[6]

[highly folded membrane]

1. highly folded chloroplast thylakoid membrane and inner mitochondrial membrane ;
 ↑ surface area for proteins to be embedded ;
2. ATP synthase: ↑ rate of diffusion of H^+ through ATP synthase
 ADP is phosphorylated to form ATP via chemiosmosis ;
3. electron carriers: ↑ rate of electron transfer ➡ energy released to pump H^+
4. photosystems embedded in thylakoid membrane: ↑ absorption of light
 ↑ rate of photoactivation ➡ ↑ rate of light-dependent reactions ;

[impermeable to H^+ ions]

5. membranes have hydrophobic core made out of nonpolar fatty acid tails
 ∴ H^+ ions are repelled / membrane is impermeable to H^+ ;
6. allow accumulation H^+ in thylakoid space and intermembrane space respectively
 to generate proton gradient (and proton motive force) ;

[permeable to small molecules]

7. membrane allows small nonpolar molecules to pass through freely ;
8. e.g. of a small molecule and its role in respiration (O_2 as final electron acceptor in oxidative phosphorylation)
9. e.g. of a small molecule and its role in photosynthesis (CO_2 required for carbon fixation in chloroplast)

[compartmentalisation]

10. e.g. of a compartmentalisation function in chloroplast (outer membrane of chloroplast compartmentalises the organelle from the cell's cytosol / chloroplast membranes compartmentalises the stroma to provide optimal conditions for enzymes involved in Calvin cycle) ;
11. e.g. of a compartmentalisation function in mitochondria (outer mitochondrial membrane compartmentalises the organelle from the cell's cytosol / inner mitochondrial membrane compartmentalises the mitochondrial matrix to provide optimal conditions for enzymes involved in Krebs cycle) ;

[proteins embedded]

12. fluidity of membrane allows electron carriers in ETC to move within the membrane to allow electron transfer ;
13. Organisation of electron carriers → arrangement in sequential order / decreasing energy level for efficiency of process;
14. AVP;

QWC [1]

At least 2 e.g. for mitochondrial membrane and 2 e.g. for chloroplast membrane;

- (b) Discuss the suitability of DNA as a hereditary molecule and its significance in DNA replication.

[9]

A. Stability of DNA [Max 4]

1. Strong covalent phosphodiester bonds within the sugar-phosphate backbone of each / one DNA strand
2. DNA molecule in a double helix with two anti-parallel strands
3. Complementary base pairing via hydrogen bonds between nitrogenous bases on DNA strands in each double helix
4. Purine always pairs with pyrimidine / Adenine with Thymine, Guanine with Cytosine
5. Hydrophobic interactions between the stacked bases within each DNA molecule
6. Base pairing ensures constant width of DNA (2nm) / constant width between the 2 sugar phosphate backbones
7. Stability of molecule leads to preservation of integrity of genetic information / sequence of DNA

B. Significance of Stability of DNA to DNA Replication [max 4]

8. Each parental strand acts as a template for the synthesis of a new daughter strand
9. The specific complementary base-pairing between deoxyribonucleotides of parental strand and those of newly synthesized daughter strand
10. Nitrogenous bases like adenine pairs with thymine & guanine pairs with cytosine
11. Each parental DNA molecule gives rise to copies of 2 DNA molecules which are identical in structure and base sequence
12. Allow for accurate replication of DNA to occur
13. Transmit identical DNA molecules and identical genetic information / sequence of DNA to daughter cells
14. AVP – reference to DNA repair mechanisms using existing strand as a template to ensure wrongly synthesized sequence is corrected and thus preserving integrity of DNA sequence

QWC [1]: Clear discussion of both stability of DNA molecule and good coherent link of the features of stability of DNA to its role in DNA replication.